

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021****Subject Code:3160923****Date:30/11/2021****Subject Name:Electrical Materials****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

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| Q.1 | (a) Compare Copper and Aluminum conducting material as | 03 |
| | (b) Enlist factors affecting resistivity of material and discuss any one in brief | 04 |
| | (c) Discuss properties of high resistivity material | 07 |
| Q.2 | (a) Give characteristic of good insulating material | 03 |
| | (b) Discuss transformer oil as insulating material | 04 |
| | (c) Enlist solid insulating materials in Electrical Machines and explain any two with its characteristics | 07 |
| OR | | |
| | (c) Enlist gaseous insulating materials in Electrical Engineering and explain any two with its characteristics | 07 |
| Q.3 | (a) Discuss effect of moisture on insulating material | 03 |
| | (b) Give brief note on amorphous material | 04 |
| | (c) Explain difference between hard and soft magnetic material | 07 |
| OR | | |
| Q.3 | (a) Give classification of magnetic material and explain in brief | 03 |
| | (b) Explain hysteresis and eddy current loss | 04 |
| | (c) Give and Justify choice of magnetic material for (1)transformer core (2)Stator of DC (3) Submarine cable(3) core of CT and PT | 07 |
| Q.4 | (a) Explain effect of temperature on semiconductor | 03 |
| | (b) Explain intrinsic and extrinsic semiconductors | 04 |
| | (c) Give properties and application of following semiconductors (1) Germanium (2) Silicon (3) Silicon carbide | 07 |
| OR | | |
| Q.4 | (a) Explain P-type and N-type semiconductor material | 03 |
| | (b) Explain effect of alloying on resistivity of material | 04 |
| | (c) Classification of Insulating material based on temperature | 07 |
| Q.5 | (a) Enlist applications of semiconductor material | 03 |
| | (b) Explain hard and soft ferrites | 04 |
| | (c) Compare Type-I and Type-II superconductor | 07 |
| OR | | |
| Q.5 | (a) Discuss Nickel Iron alloys | 03 |
| | (b) Brief note on structural and refractory material | 04 |
| | (c) Properties of superconductor | 07 |
